A

CASE STUDY

ON

BUILDING A MACHINE LEARNING MODEL

UNDERTAKEN AT

LLOYDS BANKING GROUP via FORAGE

BY

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FEBRUARY, 2026

APPROACHES TO SELECTING APPROPRIATE MACHINE LEARNING ALGORITHMS

Selecting the right machine learning algorithm is crucial for building a robust predictive model. Given the complexity of customer churn prediction, where the target variable is categorical, you need to consider several factors that influence the choice of the model. Here are some approaches to help guide your selection.

Understanding the problem type and data characteristics

In churn prediction, you're dealing with a **binary classification problem**. Key considerations include:

- **Imbalance in the data set:** Customer churn data sets often have an imbalance, where the number of churned customers is significantly less than non-churned. Techniques like **resampling**, **SMOTE (synthetic minority over-sampling technique)**, or **adjusted class weights** in algorithms are crucial for handling this imbalance effectively.
- **Feature engineering:** Advanced feature engineering techniques, such as **interaction terms**, **polynomial features**, and **dimensionality reduction (e.g., principal component analysis)**, can significantly influence the performance of algorithms, especially those sensitive to multicollinearity and high-dimensional spaces, like logistic regression and support vector machines (SVMs).

Algorithm selection and considerations

- **Logistic regression:** Preferred for its simplicity and interpretability, logistic regression can be enhanced with **regularization techniques (L1, L2)** to prevent overfitting, especially in high-dimensional data sets.
- **Decision trees and random forests:** These are powerful for capturing non-linear relationships and interactions between features. Random forests, an ensemble of decision trees, provide robustness against overfitting and allow for **feature importance analysis**, which can be crucial in understanding which factors contribute most to churn.

- **SVMs:** Effective in high-dimensional spaces and when the decision boundary is not linear. The use of **kernel tricks (e.g., RBF, polynomial)** allows SVMs to handle non-linear relationships, but they require careful tuning of hyperparameters such as **C (regularization parameter)** and **gamma**.
- **Neural networks:** While potentially offering high accuracy, especially with complex data patterns, they require large amounts of data and computational power. Techniques like **dropout**, **batch normalization**, and **early stopping** are essential to prevent overfitting.

Model evaluation and tuning

- **Cross-validation:** Advanced cross-validation techniques, such as **stratified k-fold**, ensure that each fold has a representative distribution of the target class, crucial for imbalanced data sets.
- **Hyperparameter tuning:** Employ **grid search** or **random search** for systematic exploration of the hyperparameter space. For more efficient optimization, consider using **Bayesian optimization** or **automated machine learning (AutoML)** tools.

Scalability and practical considerations

- **Model deployment:** Consider the model's scalability and integration into the business workflow. This includes **real-time prediction capabilities**, ease of updating the model with new data, and computational efficiency.
- **Interpretability vs. accuracy trade-offs:** In practice, balancing interpretability with predictive power is often necessary, especially when model decisions need to be transparent to stakeholders.

By delving into these advanced considerations, you'll be better equipped to select and fine-tune machine learning algorithms that are both accurate and aligned with the practical needs of the business context in which they will be deployed.

APPROACHES TO SELECTING AND BUILDING MACHINE LEARNING MODEL FOR CLASSIFICATION TASKS

Building a machine learning model for classification tasks, such as predicting customer churn, requires a deep understanding of both the algorithmic foundation and the practical nuances of implementation. Here are some advanced approaches to guide you through this process.

Feature selection and engineering

- **Dimensionality reduction:** Techniques such as **principal component analysis (PCA)** or **t-distributed stochastic neighbor embedding (t-SNE)** can be used to reduce the feature space, mitigating the curse of dimensionality and enhancing model performance.
- **Feature importance analysis:** Algorithms like random forests provide intrinsic measures of feature importance, which can guide the selection of the most predictive features. This step is crucial for simplifying the model and improving interpretability without sacrificing accuracy.
- **Interaction terms and polynomial features:** Introducing interaction terms and polynomial features can capture non-linear relationships between variables, which are often missed in linear models. This is particularly useful in models like logistic regression, where extending the feature space can significantly enhance predictive capability.

Model selection and evaluation

Choosing the right model involves balancing several factors:

- **Algorithm suitability:** While logistic regression and decision trees offer simplicity and interpretability, they may lack the predictive power of more complex models like **gradient boosting machines (GBMs)**, **XGBoost**, or **neural networks**. The choice often depends on the trade-off between model performance and explainability.

- **Model evaluation metrics:** In the context of imbalanced data sets, traditional metrics like accuracy are often misleading. Use metrics such as **precision, recall, F1-score,** and **ROC-AUC** to get a more accurate picture of model performance. Additionally, the **confusion matrix** provides detailed insights into the true positives, false positives, true negatives, and false negatives, which are critical for understanding model behavior.

Advanced model tuning techniques

Optimizing model performance involves fine-tuning hyperparameters:

- **Grid search and random search:** These methods are standard for hyperparameter optimization but can be computationally expensive. Grid search is exhaustive, covering all combinations of specified hyperparameters, while random search samples a wide range but in a more computationally efficient manner.
- **Bayesian optimization:** For more efficient hyperparameter tuning, Bayesian optimization offers a probabilistic approach to finding the optimal parameters, often outperforming traditional methods in terms of both accuracy and computational cost.
- **Cross-validation:** Use **stratified k-fold cross-validation** to ensure that each fold has the same proportion of classes as the original data set, which is crucial for imbalanced classification tasks. This approach helps in validating that the model generalizes well to unseen data.

Model implementation and scalability

Once a model is selected and tuned, consider its deployment and scalability:

- **Pipeline integration:** Incorporate the model into a robust data pipeline, ensuring it can handle real-time data streams and integrate seamlessly with existing systems. This includes automating data preprocessing, model prediction, and output generation.

- **Model monitoring and maintenance:** Post-deployment, continuously monitor model performance to detect drifts in data distribution or declines in accuracy. Implementing **version control** for models, along with retraining

  strategies, ensures the model remains accurate and relevant as new data becomes available.

By integrating these advanced techniques, you'll build a classification model that is not only accurate and robust but also scalable and maintainable, ensuring long-term value for the business.

TECHNIQUES FOR SUGGESTING WAYS TO EVALUATE AND MEASUER THE MODEL'S PERFORMANCE

Evaluating and measuring the performance of a machine learning model, especially in classification tasks like predicting customer churn, is crucial for understanding its effectiveness and reliability. Here are some advanced techniques and metrics to ensure comprehensive evaluation.

Choosing the right evaluation metrics

Selecting appropriate metrics depends on the specific characteristics of the data set and the business objectives:

- **Precision and recall:** These metrics are particularly important in imbalanced data sets where false positives and false negatives carry different costs. **Precision** measures the proportion of true positive predictions among all positive predictions, while **recall** measures the proportion of true positives identified out of all actual positives.
- **F1 score:** The F1 score balances precision and recall, offering a single metric that accounts for both false positives and false negatives. This is particularly useful when the costs of these errors are similar.
- **ROC-AUC (receiver operating characteristic - area under curve):** The ROC-AUC score evaluates the trade-off between true positive rates and false positive rates across different threshold settings. A higher AUC indicates better model performance across various decision thresholds.
- **Confusion matrix:** This matrix offers a detailed breakdown of the true positives, false positives, true negatives, and false negatives. It is a fundamental tool for understanding model performance, especially in terms of misclassification types.

Model calibration and validation

- **Calibration curves:** To assess how well predicted probabilities align with actual outcomes, use calibration curves. These curves compare predicted probabilities with actual outcome frequencies, helping to adjust the model to improve probability estimation.
- **Cross-validation:** Beyond simple training-validation splits, **k-fold cross-validation** ensures that the model's performance is consistently evaluated across different subsets of the data. This technique reduces the likelihood of overfitting and ensures that the model generalizes well to unseen data.
- **Bootstrapping:** This statistical method involves repeatedly resampling the data set with replacement to estimate the distribution of model performance metrics. Bootstrapping provides insights into the variability and robustness of the model's predictions.

Post-model analysis

- **Feature importance and SHAP values:** Understanding why a model makes certain predictions is crucial, especially in business contexts where decisions must be justified. **Feature importance** metrics in models like random forests and **SHAP (SHapley Additive exPlanations) values** provide insights into how each feature contributes to the model’s decisions.
- **Error analysis:** Conduct a thorough analysis of the model's errors, focusing on cases where the model performs poorly. This analysis can reveal data patterns that the model misses, leading to insights for further feature engineering or model adjustments.
- **Business impact analysis:** Beyond statistical metrics, evaluate the model's performance in terms of business outcomes. For example, measure the impact of the model on customer retention rates or revenue. This analysis helps assess the model's practical value.

Continuous monitoring and reassessment

- **Model drift detection:** Implement systems to detect model drift, which occurs when the data distribution changes over time, leading to a decline in

model performance. Techniques like monitoring prediction probabilities or feature distributions can help the early detection of drift.

- **Retraining strategies:** Based on performance monitoring, establish criteria for retraining the model. This could involve periodic retraining or retraining triggered by specific performance thresholds or detected drifts.

By employing these advanced techniques for model evaluation and measurement, you ensure that the predictive model not only performs well statistically but also aligns with business goals, providing actionable insights and reliable predictions.

CRAFTING THE PREDICTIVE BLUEPRINT

As you delve deeper into the intricacies of data science, Task 2 offers a critical opportunity to influence the strategic direction of Lloyds Banking Group. Your work in this phase will pivot from EDA to the construction of a predictive model that could significantly shape the bank's customer retention strategies.

The urgency and importance of this task cannot be overstated. Lloyds has seen a subtle but concerning trend in customer attrition, which, if not addressed, could have substantial implications for its market standing and profitability. This project is a real-world application where your findings could directly influence business decisions and outcomes.

Team collaboration and strategic impact

Working closely with Li and the Data Science & Analytics team, you are stepping into a role where your technical expertise merges with strategic business needs. The team has identified key areas where predictive insights could allow for proactive interventions, thus reducing churn and enhancing customer loyalty. Li underscores the strategic importance of your task, noting, "Our ability to predict churn allows us to personalize our engagement strategies, tailoring our approach to meet the needs and preferences of our customers. This is critical for maintaining a competitive edge."

Integrating analytical insights into business strategies

Your task involves selecting the most appropriate machine learning algorithm, which balances predictive accuracy with interpretability. This balance is crucial; while complex models may offer high accuracy, they can be challenging to explain to non-technical stakeholders who need to trust and act on these insights. Therefore, the choice of algorithm must consider not only the statistical performance but also the business context and usability.

Building the model is where your analytical skills will shine. Using the preprocessed data from Task 1, you'll train the model to identify patterns indicative of potential churn. This involves iterating on various models, tuning

hyperparameters, and validating the model to ensure it generalizes well to new data.

Evaluating and communicating model performance

Beyond building the model, it's essential to suggest robust evaluation metrics. The goal is to provide a comprehensive view of the model's performance, highlighting its strengths and identifying any limitations. Metrics like precision, recall, and F1 score will be key, especially in the context of imbalanced data sets where simple accuracy might be misleading. Furthermore, explaining the model's predictions through feature importance or other interpretability tools will be vital for aligning the model's outputs with business decisions.

Delivering actionable insights

The culmination of your work will be a detailed report. This report should present the technical aspects of the model and translate these findings into actionable business insights. Your ability to communicate complex data science concepts in a clear and actionable manner will be crucial in ensuring that the strategic implications of your work are understood and implemented by the business.

This task is a critical component of Lloyds' broader strategy to harness data-driven insights for business growth. Your contributions will play a pivotal role in shaping how the bank understands and responds to customer needs, ultimately driving customer satisfaction and loyalty. As you embark on this task, remember that your work has the potential to make a significant impact, both analytically and strategically.

TASK INSTRUCTIONS

Introduction

In this task, you will focus on developing a robust machine learning model to predict customer churn. Your objective is to select an appropriate algorithm, train and validate the model, and propose evaluation metrics that will help assess its performance. This task is pivotal for providing actionable insights that can inform business strategies at Lloyds Banking Group.

Instructions

Select an appropriate machine learning algorithm:

- Review the characteristics of the data set and the nature of the churn prediction problem.
- Consider algorithms such as logistic regression, decision trees, random forests, gradient boosting machines, or neural networks.
- Choose an algorithm that balances accuracy and interpretability, suitable for the business context.

Build and train the model:

- Use the preprocessed data set from Task 1 to train your chosen model.
- Implement techniques like cross-validation to ensure the model generalizes well to unseen data.
- Perform hyperparameter tuning to optimize the model’s performance.

Evaluate model performance:

- Select appropriate metrics to evaluate the model's performance, such as precision, recall, F1 score, ROC-AUC, and confusion matrix analysis.
- Consider the implications of each metric in the context of imbalanced data sets, ensuring that the evaluation provides a comprehensive view of the model's effectiveness.

Suggest ways to improve and utilize the model:

- Provide recommendations on how the model can be used by the business to identify at-risk customers and develop retention strategies.
- Discuss any potential improvements or adjustments to the model that could enhance its accuracy or applicability in different business scenarios.

Deliverable:

- **Report submission:** Compile a comprehensive report that includes:
    - A detailed description of the selected algorithm and the rationale behind its choice.
    - The trained model, along with performance metrics and evaluation results.
    - Suggested ways to utilize the model's predictions for business decision-making and potential areas for improvement.